

PROJECT PROPOSAL – (Only Components)

Real time Production monitoring scaling for E – 103 and other press machines

1. Project Overview

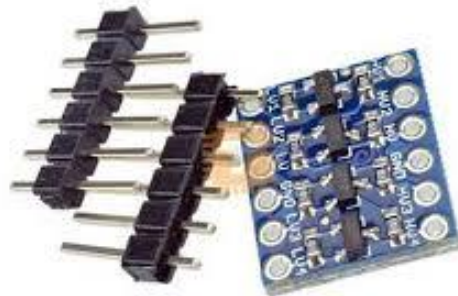
This project aims to expand the real time monitoring system, which I have already done to the E-25 press machine for 2 machines. Here I will only include the relevant components with the pictures to purchase.

*Only included components for two machines only.

01. Esp32 Microcontroller × 01



02. Logic Level Shifter × 02



03. Phone charger Adapter × 01



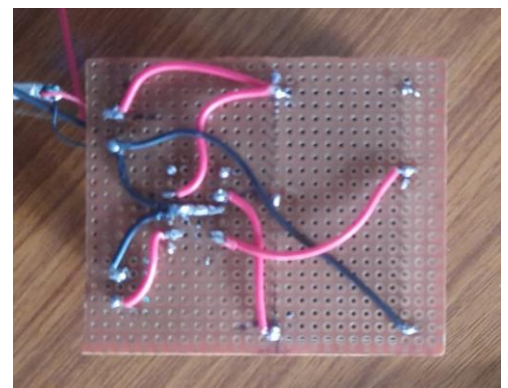
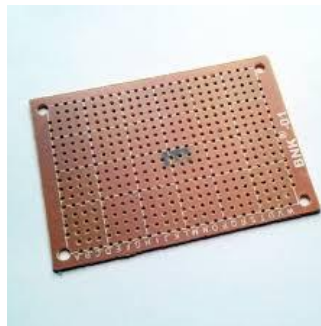
04. USB cables (only power cable required) × 03



05. Green terminal connectors 02 pin × 02



06. Dot Board (Specially dot board dots should not be interconnected.) × 02



07. Enclosure Box (1000×700 mm) × 01

- Length: 10cm
- Width: 7cm
- Depth: 5cm



08. Female headers 40 *pin* × 02



**09. Circuit Wires DC (Two colours, Ex: red and black)
(3m for each colour)**



10. NPN NO Inductive Proximity Sensor DC 6-36V LJ8A3-2-Z/BX

- Supply voltage: 6 – 36V DC
- Signal voltage: 6 – 36V
- Maximum signal current (sink): 200mA
- Measuring range: 0 – 2mm
- Output type: NPN NO (normally open)
- Internal pull-up resistance: 10k Ω
- Measuring method: Inductive
- Cable length: ~1m
- Thread: M8
- Dimensions: 8 x 8 x 45mm



11. SN04-N PNP NO Inductive Proximity Sensor

- SN04-P (PNP NO)
- Dimensions: 18*18*36 mm
- Sensing distance: 4 mm
- Test objects: metal (copper · iron · aluminum · gold)
- Working voltage: DC 6 - 30VDC
- Output type: PNP NO
- Output current: 300 MA
- Case material: plastic ABS



Approximated Budget

No	Item	Quantity	Unit Cost (LKR)	Cost (LKR)
01	ESP32 Microcontroller	01	1650	1650
02	AC-DC Power Adapter 230V to 5V	01	650	650
03	1000×700 mm Industrial IP box with hanging	01	600	600
04	Dot Board (Specially check)	02	100	200
05	USB Cable – (Power only)/charging cable	03	350	1050
06	DC Circuit Wires – (20–24 AWG) / two colour	3m for each colour	75	150
07	Female Pin Headers (40 pin)	02	50	100
08	Logic Level Shifter	02	125	250
09	Green Terminal Connectors × 2	02	20	40
10	Inductive Proximity Sensor (NO, NPN) LJ8A3-2-Z/BX	01	1000	1000
12	SN04-N PNP NO Inductive Proximity Sensor	01	740	740
11	Total ($\approx$ <i>Approximated</i>)	-	-	6430